

Documentation for BigQuery Fintech Dataset

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1 Preliminary Analysis of the Dataset

For this project, we make use of the BigQuery Fintech Dataset that is publicly available at <https://www.kaggle.com/dataset/fintech-dataset/data?select=loan.csv>. We first analyse the `loans.csv` dataset.

1.1 `loans.csv` Dataset

The dataset consists of 270299 rows and 18 columns occupying space of 172MB. A following is the list of the columns together with their datatypes:

Column	dtype
<code>loan_id</code>	<code>int64</code>
<code>customer_id</code>	<code>str</code>
<code>loan_status</code>	<code>str</code>
<code>loan_amount</code>	<code>float64</code>
<code>state</code>	<code>str</code>
<code>funded_amount</code>	<code>float64</code>
<code>term</code>	<code>str</code>
<code>int_rate</code>	<code>float64</code>
<code>installment</code>	<code>float64</code>
<code>grade</code>	<code>str</code>
<code>issue_d</code>	<code>datetime64[us]</code> [†]
<code>issue_date</code>	<code>datetime64[us]</code> [†]
<code>issue_year</code>	<code>float64</code>
<code>pymnt_plan</code>	<code>bool</code>
<code>type</code>	<code>str</code>
<code>purpose</code>	<code>str</code>
<code>description</code>	<code>str</code>
<code>notes</code>	<code>str</code>

Table 1: Column analysis of `loans.csv`. Dtypes are those inferred by `pandas.read_csv` (pandas 3.0.3, in which text columns take the dedicated `str` dtype). [†]`issue_d` and `issue_date` are stored as text (`Jun-13` and `June 2013` respectively) and are read as `str`; the dtype shown is the result of parsing them with `pandas.to_datetime`.

The dataframe was free of missing values including placeholders for all columns except `description`. Upon inspection of the columns, it is noted that the following pairs of columns are duplicated: `loan_amount`, `funded_amount` for the principal amount, `issue_date`, `issue_d` for the date of issue, `purpose`, `description` for the purpose of loan. Furthermore, the `pymnt_plan` nearly constant and consisting of 99.99% of `False` variables.

There were also type issues noted in the `type` column, entries like ‘individual’ and ‘INDIVIDUAL’ counted as unique entries in pandas and were subsequently merged into single entries.

To handle all the data quality issues identified, a new dataset called ‘`loan_cleaned.csv`’ was created by dropping the duplicate columns (`funded_amount`, `issue_d`, `description`), the non-varying columns

(`notes`, `pymnt_plan`) and the identifier columns (`loan_id`, `customer_id`).¹, and fixing the entry inconsistencies seen in the `type` column.

¹This step will be reversed in the future to allow for joining with the other dataset `customer.csv`.